Part II Particle & Nuclear Physics — Cheat Sheet

Units, Constants & Kinematics

Natural units: $\hbar = c = 1$. $\hbar c = 197\,\text{MeV·fm}$; $\hbar = 6.6\times10^{-25}\,\text{GeV·s}$; $1\,\text{b} = 10^{-28}\,\text{m}^2$; $1\,\text{u} = 931.5$ MeV.

$\alpha = \dfrac{e^2}{4\pi\epsilon_0\hbar c} \to \dfrac{e^2}{4\pi} \approx \dfrac{1}{137}$, so $e \approx 0.30$.

$$E^2 = p^2 + m^2, \quad E = \gamma m,\ |\vec p| = \gamma\beta m$$

$$\beta = \tfrac{|\vec p|}{E}, \quad T = E - m = (\gamma - 1)m$$

4-vector: $p^\mu p_\mu = E^2 - |\vec p|^2 = m^2$ (invariant mass).

$$M_X^2 = \left(\sum E_i\right)^2 - \left(\sum \vec p_i\right)^2$$

Two-body: $M_X^2 = m_1^2 + m_2^2 + 2(E_1E_2 - |\vec p_1||\vec p_2|\cos\theta)$.

Centre-of-mass energy $\sqrt{s}$

Fixed target ($E_1 \gg m$): $s \approx 2E_1 m_2$. Symmetric collider: $\sqrt{s} = 2E$.

Decays, Rates & Cross-sections

$$N(t) = N_0 e^{-\lambda t}, \quad A = \left|\tfrac{dN}{dt}\right| = \lambda N$$

$\tau = 1/\lambda$; $\tau_{1/2} = \dfrac{\ln 2}{\lambda} = 0.693\,\tau$.

Chain $N_1 \to N_2 \to N_3$: $\dfrac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2$.

Fermi's Golden Rule:

$$\Gamma = 2\pi|M_{fi}|^2\rho(E_f)$$

Width $\Gamma = 1/\tau = \lambda$ (n.u.). Partial widths: $\Gamma = \sum_f \Gamma_f$; branching $B_f = \Gamma_f/\Gamma$, $\sum B_f = 1$.

Cross-section: $\sigma = \Gamma/\Phi$; flux $\Phi = v_i/L^3$.

Attenuation: $N = N_0 e^{-\sigma n L}$; mean free path $= 1/(n\sigma)$.

Born approximation

$$\frac{d\sigma}{d\Omega} = \frac{E^2}{(2\pi)^2}\left|\int e^{-i\vec q\cdot\vec r}V(\vec r)\,d^3\vec r\right|^2$$

Rutherford ($V = -Z\alpha/r$):

$$\frac{d\sigma}{d\Omega} = \frac{Z^2\alpha^2}{4E^2\sin^4(\theta/2)}, \quad q^2 = -4E_iE_f\sin^2\tfrac{\theta}{2}$$

Resonances (Breit–Wigner)

$$\sigma = \frac{\pi g}{p_i^2}\,\frac{\Gamma_{Z\to i}\Gamma_{Z\to f}}{(E-E_0)^2 + \Gamma^2/4}$$

$g = \dfrac{2J_Z + 1}{(2J_a+1)(2J_b+1)}$ (spin factor).

Peak: $\sigma_{\text{peak}} = \dfrac{4\pi g\,\Gamma_i\Gamma_f}{p_i^2\Gamma^2}$. With $B_i = \Gamma_i/\Gamma$: $\sigma_{\text{el}} = \dfrac{4\pi g B_i^2}{p_i^2}$, $\sigma_{\text{tot}} = \dfrac{4\pi g B_i}{p_i^2}$.

Standard Model & Exchange

12 fermions (spin ½) + 5 spin-1 bosons + Higgs (spin 0).

Klein–Gordon (bosons): $\dfrac{\partial^2\psi}{\partial t^2} = (\nabla^2 - m^2)\psi$, $E = \pm\sqrt{p^2+m^2}$ ⇒ antimatter. Dirac eqn for fermions.

Yukawa potential:

$$V(r) = -\frac{g^2}{4\pi}\frac{e^{-mr}}{r}, \quad \text{range} \sim \tfrac{1}{m}$$

Virtual particle off-shell mass $q^2 = E^2 - |\vec p|^2 \neq m^2$.

Propagator: $M_{fi} = \dfrac{g^2}{q^2 - m^2}$; massless: $M_{fi} = g^2/q^2$.

Feynman Rules

- Vertex → coupling g. EM: $g = Qe$; weak: g_W; strong: $\sqrt{\alpha_s}$.
- Internal line → propagator $1/(q^2 - m^2)$.
- Conserve $E, \vec p, J, Q$, lepton#, baryon# $B = \frac{1}{3}(n_q - n_{\bar q})$, strangeness at every vertex.
- Total amplitude $M = M_1 + M_2 + \ldots$; rate $\propto |M|^2$.
- Channels: $s = (p_1+p_2)^2$, $t = (p_1 - p_3)^2$, $u = (p_1 - p_4)^2$.

Colliders & Detectors

Bending: $B = \dfrac{p[\text{GeV}]}{0.3\,r[\text{m}]}$. Synchrotron loss/orbit $\propto E^4/(m^4 r)$.

Tracker: $p = 0.3B\dfrac{L^2}{8s}$ (sagitta s); $\dfrac{\sigma_p}{p} \propto p$.

Calorimeter: $\dfrac{\sigma_E}{E} \propto \dfrac{1}{\sqrt{E}}$.

QED

EM vertex coupling $\propto Q\sqrt{\alpha}$, $\alpha = e^2/4\pi$. Never changes flavour.

$M = \dfrac{e^2}{q^2} = \dfrac{4\pi\alpha}{q^2}$. "Spinless" e^-p: $\dfrac{d\sigma}{d\Omega} = \dfrac{4\alpha^2E^2}{q^4}$.

$e^+e^- \to \mu^+\mu^-$ (with spin):

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4s}(1+\cos^2\theta), \quad \sigma = \frac{4\pi\alpha^2}{3s}$$

Running: $\alpha(0) \approx \frac{1}{137}$, $\alpha((100\,\text{GeV})^2) \approx \frac{1}{128}$ (increases with q^2).

QCD

$\alpha_s = \dfrac{g_s^2}{4\pi} \sim 1$. Colour r, g, b; 8 gluons (carry colour, self-interact).

Colour singlets: meson $\frac{1}{\sqrt{3}}(r\bar r + g\bar g + b\bar b)$; baryon $\frac{1}{\sqrt{6}}(rgb + \ldots)$.

QCD potential:

$$V_{QCD} = -\frac{4}{3}\frac{\alpha_s}{r} + kr, \quad k \approx 1\,\text{GeV/fm}$$

$C = 4/3$ ($q\bar q$ meson), $2/3$ (baryon). Confinement ⇒ jets, hadronisation. Asymptotic freedom: α_s small at high E.

Ratio R:

$$R = \frac{\sigma(e^+e^- \to \text{had})}{\sigma(e^+e^- \to \mu^+\mu^-)} = 3\sum_i Q_i^2$$

Steps: uds → 2, +c → 10/3, +b → 11/3. With gluon: $R = 3\sum Q_q^2(1 + \alpha_s/\pi)$. 3-jet rate $\propto \alpha_s$.

Quark Model of Hadrons

$\psi = \psi_{\text{space}}\psi_{\text{flavour}}\psi_{\text{spin}}\psi_{\text{colour}}$. Mesons $q\bar q$, baryons qqq.

Parity: $P = P_1P_2(-1)^L$; $q\bar q$ meson ($L=0$): $P = -1$. $q/\ell: +1$, $\bar q/\bar\ell: -1$, $\gamma, g, W, Z: 1^-$.

Meson mass (L=0)

$$M_{q\bar q} = m_1 + m_2 + A\frac{\vec S_1\cdot\vec S_2}{m_1m_2}$$

0^-: $-\frac{3A}{4m_1m_2}$; 1^-: $+\frac{A}{4m_1m_2}$. ($\vec S_1\cdot\vec S_2 = -\frac{3}{4}$ or $+\frac{1}{4}$.)

Baryon mass (L=0)

$$M_{qqq} = \sum m_i + A\sum_{i<j}\frac{\vec S_i\cdot\vec S_j}{m_im_j}$$

$\sum_{i<j}\vec S_i\cdot\vec S_j = -\frac{3}{4}$ $(J = \frac{1}{2})$ or $+\frac{3}{4}$ $(J = \frac{3}{2})$.

Magnetic moments

$\mu_q = \dfrac{q_q\hbar}{2m_q}$; $\mu_B = \dfrac{e\hbar}{2m_e}$, $\mu_N = \dfrac{e\hbar}{2m_p}$. Dirac: $g_s = 2$.

$p = \frac{4}{3}\mu_u - \frac{1}{3}\mu_d$, $n = \frac{4}{3}\mu_d - \frac{1}{3}\mu_u$ ⇒ $\mu_n/\mu_p = -\frac{2}{3}$.

Heavy: charmonium $c\bar c$, bottomonium $b\bar b$; $J/\psi\ 1^-$; Zweig rule ⇒ narrow widths.

Weak Force (CC)

$W^\pm$, $m_W = 80.4$ GeV; flavour-changing & parity-violating. Couples to LH particles / RH antiparticles.

Helicity $h = \dfrac{\vec S\cdot\vec p}{|\vec p|}$. Coupling prob $\frac{1}{2}(1 \mp v/c)$ particles.

Fermi theory:

$$\frac{g_W^2}{8m_W^2} = \frac{G_F}{\sqrt{2}}, \quad G_F = 1.166\times10^{-5}\,\text{GeV}^{-2}$$

$\alpha_W = g_W^2/4\pi \approx \frac{1}{30}$ (intrinsically stronger than EM). νn: $\sigma = G_F^2 s/\pi$.

Muon decay (Sargent): $\Gamma_\mu = \dfrac{G_F^2 m_\mu^5}{192\pi^3}$. Lepton universality.

Quark mixing (CKM)

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = V_{CKM}\begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

$d' = d\cos\theta_C + s\sin\theta_C$, $\theta_C \approx 13°$. Same family = Cabibbo allowed; cross = suppressed.

Electroweak (GWS)

$Z = W_3\cos\theta_W - B\sin\theta_W$, $A = W_3\sin\theta_W + B\cos\theta_W$.

$$g_W = \frac{e}{\sin\theta_W}, \ g_Z = \frac{g_W}{\cos\theta_W}, \ m_Z = \frac{m_W}{\cos\theta_W}$$

$\sin^2\theta_W \approx 0.231$. Z coupling $\propto (I_3 - Q\sin^2\theta_W)$.

Z resonance:

$$\sigma = \frac{3\pi}{s}\frac{\Gamma_{ee}\Gamma_{f\bar f}}{(\sqrt{s} - m_Z)^2 + \Gamma_Z^2/4}; \ \sigma_{\text{pk}} = \frac{12\pi}{m_Z^2}\frac{\Gamma_{ee}\Gamma_f}{\Gamma_Z^2}$$

$m_Z = 91.19$, $\Gamma_Z = 2.495$ GeV. $N_\nu = \Gamma_{\nu\nu}^{\text{meas}}/\Gamma_{\nu\nu}^{\text{SM}} \approx 3$.

W: $\Gamma(W \to e\nu) = \dfrac{g_W^2 m_W}{48\pi} = \dfrac{G_F m_W^3}{6\sqrt{2}\pi}$; $B(W \to q\bar q) = \frac{2}{3}$.

Higgs & Neutrinos

Higgs $m_H \approx 125$ GeV, $J^P = 0^+$; couplings $\propto$ mass. $V(\phi) = a\phi^4 - b\phi^2$, $|\phi_{GS}^2| = b/2a$.

Top: $\tau_t \sim 10^{-25}$s, $V_{tb} \approx 1 \Rightarrow t \to Wb$ before hadronising.

Neutrino oscillations

$$P(\nu_\mu \to \nu_e) = \sin^2 2\theta\,\sin^2\left(\frac{\Delta m^2 L}{4E}\right)$$

$$P = \sin^2 2\theta\,\sin^2\left(\frac{1.27\,\Delta m^2[\text{eV}^2]\,L[\text{km}]}{E_\nu[\text{GeV}]}\right)$$

Implies $m_\nu \neq 0$ (BSM). Cherenkov: $\cos\theta_C = 1/(n\beta)$.

Nuclear Properties & SEMF

$m(A, Z) = Zm_p + (A - Z)m_n - B$.

Semi-empirical mass formula:

$$B = a_V A - a_S A^{2/3} - \frac{a_C Z^2}{A^{1/3}} - \frac{a_A(N-Z)^2}{A} + \delta$$

$a_V = 15.8$, $a_S = 18.0$, $a_C = 0.72$, $a_A = 23.5$, $a_P = 33.5$ MeV.

$\delta = +a_P A^{-3/4}$ (e-e), $-a_P A^{-3/4}$ (o-o), 0 (e-o).
Most stable Z: $\partial m/\partial Z|_A = 0$.

Size & shape

Form factor $F(q^2) = \int \rho(r) e^{i\vec{q}\cdot\vec{r}} d^3r$; $\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega}\right)_{\text{pt}} |F(q^2)|^2$.

$\rho(r) = \dfrac{\rho(0)}{1+e^{(r-R)/s}}$ (Fermi), $R = r_0 A^{1/3}$, $r_0 \approx 1.2$ fm.

Mirror nuclei: $\Delta E_C = \frac{3}{5}\frac{A\alpha}{R}$. Quadrupole $Q = \frac{1}{e}\int(3z^2 - r^2)\rho\, d^3r$.

Magnetic: $\vec{\mu} = \frac{\mu_N}{\hbar}(g_L \vec{L} + g_s \vec{s})$.

Shell Model & Excited States

Magic: 2, 8, 20, 28, 50, 82, 126. Saxon–Woods $V = \frac{-V_0}{1+e^{(r-R)/s}}$ + spin-orbit.

$\vec{L}\cdot\vec{S} = \frac{1}{2}[j(j+1) - L(L+1) - s(s+1)]$; splitting $\Delta E = \frac{1}{2}(2L+1)V_{so}$ (larger j lower).

J^P: e-e $\to 0^+$; odd-A $\to$ unpaired nucleon, $P = (-1)^L$.

Schmidt: $g_J = g_L \pm \dfrac{g_s - g_L}{2L+1}$, $j = L \pm \frac{1}{2}$. p: $g_L = 1, g_s = 5.586$; n: $g_L = 0, g_s = -3.826$.

Rotational: $E = \dfrac{\hbar^2}{2I_{\text{eff}}} J(J+1)$, $\dfrac{E(4^+)}{E(2^+)} = \frac{10}{3}$.
Vibrational ratio ≈ 2.

Nuclear Decay: α, β, γ

α decay ($A \gtrsim 150$–210)

$Q = m_X - m_Y - m_{He} > 0$. Gamow tunnelling:

$$P = e^{-2G},\ G = \int_R^{R'} \sqrt{\tfrac{2m_\alpha(V(r)-E_0)}{\hbar^2}}\, dr$$

$\lambda = fP$, $f \approx v/2R$. Geiger–Nuttall: $\ln\lambda \approx -Z'/\sqrt{E_0}$+const.

β decay (weak, parity-violated)

β^-: $E_0 = M_X - M_Y$; β^+: $E_0 = M_X - M_Y - 2m_e$ (atomic).

$$\Gamma = \frac{G_F^2 |M_{\text{nuc}}|^2}{2\pi^3} \int_0^{E_0} (E_0 - E_e)^2 p_e^2 F(Z, E_e)\, dE_e$$

Sargent $\Gamma \propto E_0^5$. Kurie: $\sqrt{\frac{d\Gamma/dp_e}{p_e^2}} \propto (E_0 - E_e)$.
Fermi ($S_{e\nu} = 0$, $\Delta J = 0$) / Gamow–Teller ($S_{e\nu} = 1$, $\Delta J = 0, \pm1$, no $0 \to 0$). Comparative $f\tau_{1/2}$.

γ decay

$\Gamma(E1) \approx \frac{4}{3}\alpha E_\gamma^3 R^2$. EL: $P = (-1)^L$; ML: $P = (-1)^{L+1}$. $0 \to 0$ forbidden. Rate falls $\sim 10^{-3}$ per multipole.

Fission & Fusion

Most stable at $A \sim 60$. Spontaneous fission if $E_0 = B_1 + B_2 - B > 0$; symmetric ($y = \frac{1}{2}$) max $E_0 = 0.37 a_C \frac{Z^2}{A^{1/3}} - 0.26 a_S A^{2/3}$.

Instability ⇒ $\dfrac{Z^2}{A} > \dfrac{2a_S}{a_C} \approx 47$. Fission barrier E_f (tunnelling, c.f. α).
^{238}U: $E_0 \sim 200$ MeV ($\sim 10^6\times$ chemical).